

RENTAL_PROPERTY							
PropertyID	Address	OwnerID	OwnerName	OwnerPhone	TenantID	TenantName	Rent
P001	123 Main St	O10	Smith	555-1234	T100	Alicè	1200
P002	456 Oak Ave	O10	Smith	555-1234	T101	Bob	1500
P003	789 Elm St	O20	Jones	555-5678	T102	Charlie	1000
P001	123 Main St	O10	Smith	555-1234	T103	David	1200
P004	123 Somewhere	O20	Jones	555-5678	T100	Alice	1300

What are the functional dependencies in the above table?

PropertyID $\rightarrow$ Address, OwnerID

OwnerID $\rightarrow$ Owner Name, OwnerPhone

TenantID $\rightarrow$ TenantName, Rent

Using the closure method, determine the candidate key(s) of the above table.

1. List essential column(s)
(PropertyID, TenantID)

2. Find closure of essential column(s)

$(\text{PropertyID}, \text{TenantID})^+ = (\text{PropertyID}, \text{TenantID}, \text{Address}, \text{OwnerID}, \text{OwnerName}, \text{OwnerPhone}, \text{TenantName}, \text{Rent})$

Candidate key = (PropertyID, TenantID)

The table below stores patient, doctor, and appointment information for a small clinic. DoctorID is unique for each doctor. PatientID is unique for each patient. Because the clinic is so small, doctors can see many different patients on the same date, but patients can only see one doctor per appointment and cannot see the same doctor twice on any given day. Below is sample data stored by the clinic.

DOCTOR_PATIENT_APPOINTMENT

DoctorID	DoctorName	PatientID	PatientName	PatientPhone	AppointmentDate	AmountBilled
D1036	Jennifer Keller	P102	Samantha Carter	334-275-8191	05/28/2002	\$560
D0011	Janet Fraiser	P104	Daniel Jackson	206-342-8631	08/04/2001	\$3,000
D0011	Janet Fraiser	P104	Daniel Jackson	206-342-8631	10/12/2007	\$8,000
D0011	Janet Fraiser	P101	Jack O'Neill	717-550-1675, 484-782-7050, 484-994-9689	08/04/2001	\$560
D0124	Carolyn Lam	P102	Samantha Carter	334-275-8191	01/19/2005	\$6,782
D1032	Carson Beckett	P201	John Sheppard	248-762-0356, 336-119-4784	08/04/2001	\$200

DoctorID unique for each doctor

PatientID unique

Patients can see one doctor per appointment

Doctors can see many patients on same day

FDs

DoctorID → DoctorName

PatientID → PatientName, PatientPhone

DoctorID, PatientID, AppointmentDate → AmountBilled

Closure

$(\text{DoctorID}, \text{PatientID}, \text{AppointmentDate})^+ = (\text{DoctorID}, \text{PatientID}, \text{AppointmentDate}, \text{DoctorName}, \text{PatientName}, \text{PatientPhone}, \text{AmountBilled})$